

Working Title

YT

September 29, 2026

Abstract

Introduction
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1 Introduction

the time of symptom onset determines access to treatment. (stroke treatment guidelines)
study showed that unknown-onset stroke is common, distinct from wake-up stroke, and associated with lower reperfusion treatment, and poorer outcomes [1].

Main Aim:

The primary aim of this study was to examine temporal and geographic variation in uncertain symptom-onset stroke among stroke patients in Norway.

2 Methods

Data from the norwegian stroke registry, statistics norway

fom 2014 - tom 2013

inklusion criteria: I61 - hjerneblødning I63 - hjerneinfarkt I64 - hjerneslag, ikke spesifisert som infarkt eller blødning

nødvendig med sikre akutte fokale nevrologiske utfall

for bevistløse pasienter: akutte fokale nevrologiske utfall med positiv bildediagnostikk

innlagt innen 28 døgn etter symptomdebut (nytt slag etter 7 døgn registreres som nytt tilfelle, men ikke de innen 7 døgn)

ved trombolyse der det ikke blir sekvele eller bildefunn etter innleggels: kodes som I 63.

Pasienter med normal kontroll MR etter trombolytisk behandling skal kodes som hjerneinfarkt i ICD10.

Flow chart

88228

Innlagt i sykehus: 4571

83657

Våknet med symptom: 14757

68900

komnr missing: 38

68862

Table 1: Characteristics of patients

	Uncertain onset (n = 12,612)	Other stroke (n = 55,233)	SMD
Age, mean (SD)	73.89 (13.4)	73.31 (13.4)	0.043
Female	5,945 (47.1)	25,337 (45.9)	0.025
Stroke type, Infarction	10,813 (85.7)	47,360 (85.7)	0.009
Previous TIA	880 (7.0)	4,356 (7.9)	0.043
Previous MI	1,482 (11.8)	6,816 (12.3)	0.020
Atrial fibrillation	2,917 (23.1)	13,119 (23.8)	0.015
Diabetes	2,438 (19.3)	9,234 (16.7)	0.069
Antihypertensive medication	6,508 (51.6)	28,464 (51.5)	0.040
Lipid-lowering medication	3,700 (29.3)	16,821 (30.5)	0.050
Thrombolysis	161 (1.3)	12,148 (22.0)	0.685
Smoking status			0.086
Never smoker	4,673 (37.1)	20,966 (38.0)	
Current smoker	2,717 (21.5)	10,234 (18.5)	
Former smoker	2,998 (23.8)	14,538 (26.3)	
Unknown	2,224 (17.6)	9,495 (17.2)	
Level of Consciousness (admission)			0.087
Alert	10,856 (86.1)	47,236 (85.5)	
Mildly Drowsy	800 (6.3)	4,283 (7.8)	
Markedly Drowsy	349 (2.8)	1,617 (2.9)	
Unresponsive	432 (3.4)	1,699 (3.1)	
Unknown	175 (1.4)	398 (0.7)	
Pre-stroke modified Rankin Scale			0.098
No symptoms	6,884 (54.6)	31,923 (57.8)	
No significant disability	1,927 (15.3)	8,805 (15.9)	
Slight disability	1,369 (10.9)	5,319 (9.6)	
Moderate disability	931 (7.4)	3,829 (6.9)	
Moderately severe disability	385 (3.1)	1,612 (2.9)	
Severe disability	54 (0.4)	227 (0.4)	
Unknown	1,062 (8.4)	3,518 (6.4)	
Living situation			0.173
Living with others	6,192 (49.1)	31,614 (57.2)	
Living alone	5,701 (45.2)	20,548 (37.2)	
Institution/nursing home	376 (3.0)	1,490 (2.7)	
Unknown	333 (2.8)	1,581 (2.8)	
Education			0.055
No education	60 (0.5)	203 (0.4)	
Compulsory education	4,150 (33.2)	17,149 (31.3)	
Upper-secondary education	5,868 (46.9)	25,743 (47.0)	
Higher education	2,419 (19.3)	11,606 (21.2)	
Unknown	15 (0.1)	60 (0.1)	
Household income			0.068
Low	3,325 (26.4)	13,437 (24.3)	
M. low	3,138 (24.9)	13,278 (24.0)	
M. high	3,009 (23.9)	13,470 (24.4)	
High	3,092 (24.5)	14,885 (26.9)	
Unknown	48 (0.4)	163 (0.3)	

SMD = standardized mean difference; SD = standard deviation; TIA = transient ischaemic attack.

2.1 Statistical analysis

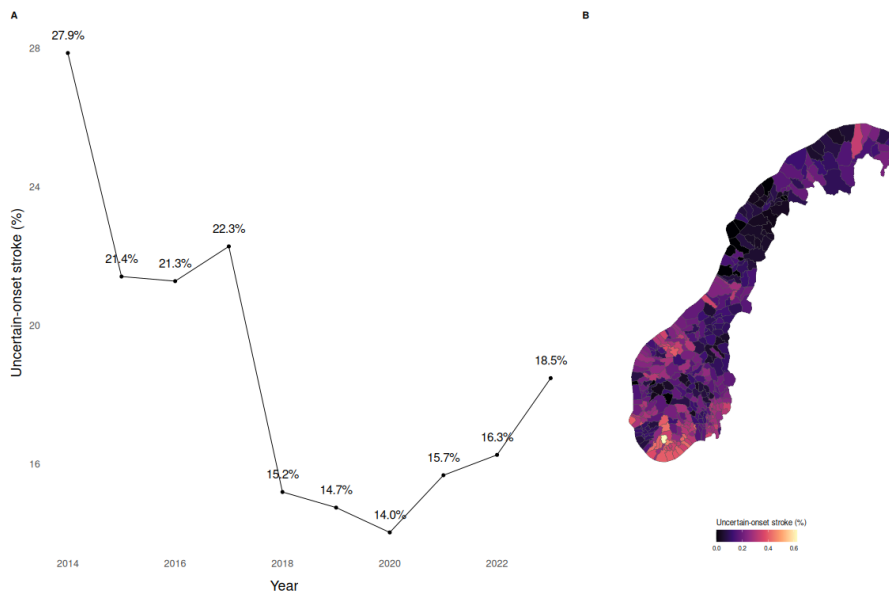
3 Results

The study population included 67,845 patients with stroke registered in the Norwegian Stroke Registry between 2014 and 2023, of whom 12,612 (18.6%) had uncertain-onset stroke (table 1). Most patient characteristics were similar between patients with uncertain-onset stroke and all other strokes, with standardized mean differences generally below 0.10. The largest differences were observed for living situation, and thrombolysis, while differences in demographic characteristics, stroke type, and cardiovascular history were modest. Living alone was more common among patients with uncertain-onset stroke (45.2% vs 37.2%), and they recieved thrombolysis less frequently (1.3% vs 22.0%).

Temporal and geographic variation in uncertain-onset stroke

The crude proportion of uncertain-onset stroke varied over time, with the heigest proportion in 2014 (27.9%) and the lowest in 2020 (14.0%). The proportion declined during the early years of the study period, with a marked reduciton between 2017 and 2018. It remained relatively low through 2020, followed by a gradual increase to 18.5% in 2023 (figure 1A).

Figure 1: Temporal and geographic variation in uncertain-onset stroke. A) Annual proportion of patients with uncertain-onset stroke, 2014–2023. B) Municipality-level geographic variation in uncertain-onset stroke.



In addition to the temporal variation, substantial geographic variation in uncertain-onset stroke was observed across municipalities, ranging from 0% to 62% with a median of 16.7% (IQR 10.2% and 25.8%) (Figure 1B). Areas with relatively high and low proportions appeared to cluster together geographically rather than an even distribution across the country.

unadjusted and unadjusted pairs from 2014 to 2023

Table 2: Fixed effect estimates with and without BYM2 spatial effect. Values are log-odds coefficients with 95% credible intervals. Bold indicates intervals excluding zero.

Variable	Without BYM2 β (95% CrI)	With BYM2 β (95% CrI)	$\Delta\beta$
(Intercept)	-1.701 (-1.924, -1.478)	-1.834 (-2.065, -1.603)	0.133
Bosituasjon: bodde alene	0.408 (0.366, 0.450)	0.428 (0.385, 0.471)	-0.020
Bosituasjon: bodde i institusjon/sykehjem	0.321 (0.216, 0.425)	0.349 (0.242, 0.457)	-0.028
Bevissthetsgrad: døsig, reagerer adekvat ved lett stimulering	-0.202 (-0.280, -0.125)	-0.174 (-0.254, -0.095)	-0.028
Bevissthetsgrad: døsig, reagerer først ved kraftigere/gjentatt stimulering	-0.079 (-0.192, 0.035)	-0.027 (-0.144, 0.089)	-0.051
Bevissthetsgrad: reagerer ikke / ikke-måltrettet bevegelse	0.073 (-0.031, 0.177)	0.128 (0.020, 0.235)	-0.055
Røyking: røyker	0.153 (0.096, 0.210)	0.148 (0.090, 0.207)	0.005
Røyking: eks-røyker	-0.041 (-0.092, 0.010)	-0.011 (-0.064, 0.042)	-0.030
Kjønn: ErMann = nei	-0.074 (-0.117, -0.031)	-0.070 (-0.115, -0.026)	-0.004
Utdanning: obligatorisk utdanning	-0.075 (-0.247, 0.097)	-0.062 (-0.239, 0.115)	-0.013
Utdanning: mellomutdanning	-0.107 (-0.278, 0.065)	-0.107 (-0.283, 0.070)	-0.000
Utdanning: universitets- og høyskoleutdanning	-0.153 (-0.330, 0.024)	-0.124 (-0.306, 0.057)	-0.029
Inntekt: samlet kvartil 2	-0.056 (-0.110, -0.002)	-0.057 (-0.113, -0.001)	0.001
Inntekt: samlet kvartil 3	-0.089 (-0.147, -0.032)	-0.081 (-0.140, -0.021)	-0.009
Inntekt: samlet kvartil 4	-0.152 (-0.217, -0.087)	-0.130 (-0.197, -0.063)	-0.022
Inntekt: samlet ukjent/9	0.125 (-0.196, 0.445)	0.141 (-0.188, 0.470)	-0.016
Tidligere hjerneslag: nei	-0.032 (-0.082, 0.017)	-0.030 (-0.081, 0.021)	-0.002
Tidligere TIA: nei	0.110 (0.041, 0.179)	0.099 (0.028, 0.170)	0.011
Tidligere hjerteinfarkt: nei	0.060 (-0.001, 0.121)	0.042 (-0.022, 0.105)	0.019
Atrieflimmer: nei	0.049 (0.002, 0.097)	0.040 (-0.009, 0.089)	0.010
Diabetes: nei	-0.213 (-0.262, -0.161)	-0.232 (-0.283, -0.181)	0.019
ASA pre: nei	0.064 (0.017, 0.111)	0.053 (0.005, 0.101)	0.011
Statin/lipid pre: nei	0.049 (0.002, 0.095)	0.065 (0.017, 0.113)	-0.016
Dipyridamol pre: nei	0.167 (0.062, 0.273)	0.131 (0.023, 0.240)	0.036
BT-medikasjon pre: nei	-0.030 (-0.072, 0.013)	-0.006 (-0.050, 0.038)	-0.023

4 Discussion

This is a placeholder Discussion section added to verify the LaTeX build and editor setup. Replace this text with interpretation of results and context.

Supplementary

References

- [1] M Søyland et al. “Wake-up stroke and unknown-onset stroke; occurrence and characteristics from the nationwide Norwegian Stroke Register”. In: *European Stroke Journal* 7.2 (2022), pp. 143–150. DOI: 10.1177/23969873221089800.

